ChatGPT

Canadian Mines – Methods and Ore Types

Gold Mines:
- **Brucejack (BC, Newmont):** An underground orogenic gold mine using long-hole stoping. Ore is crushed underground and conveyed to surface, where a fully enclosed mill (gravity + flotation + CIL) produces gold–silver doré bars and a flotation concentrate. The ore is high-grade quartz vein gold (largely free-milling, with gold and silver mostly native).
- **Canadian Malartic (QC, Agnico Eagle/Yamana):** A combined open-pit (Odyssey) and underground gold operation. Ore is processed in a ~60,000 t/d plant (two-stage crushing, grinding, gravity and carbon-in-pulp (CIP) cyanidation) with electrowinning, yielding doré. The gold-bearing greenstone ore is relatively low-sulfide (free-milling) and yields high recovery by conventional gravity/CIP methods.
- **Casa Berardi (QC, Hecla):** An open pit and underground lode gold mine. A 4,400 t/d mill (crush, SAG/ball mill + CIL) recovers gold via carbon-in-leach, electrowinning and smelting to doré [1] [2]. The ore (vein-hosted orogenic gold) is free-milling (no refractory pretreatment needed).
- **Detour Lake (ON, Kirkland Lake):** A large open-pit gold mine. Ore is processed in a conventional 13.9 Mtpa plant (crushing, SAG mill, gravity circuits, cyanidation, CIL), producing doré [3]. The Abitibi greenstone ore is free-milling, with gravity recovery (and CIP) capturing nearly all gold without special pretreatment.
- **Eagle River (ON, Wesdome):** A high-grade underground gold mine (long-hole stoping). It has a 1,200 t/d mill using gravity concentration and carbon-in-pulp (Merrill-Crowe) to recover gold to doré [4]. About 40–60% of the gold is recovered by gravity [5] (reflecting free-milling coarse gold), with the balance in cyanide-CIP; ore is primarily free-milling.
- **Éléonore (QC, Newmont):** An underground gold mine (long-hole stoping). The 7,000 t/d plant uses crushing, grinding, gravity separation and flotation followed by cyanidation [6]. About 80–90% of gold is recovered (reportedly ~50–60% by gravity) and final product includes doré and concentrates. The ore is an Archean sulfide gold deposit (pyrrhotite veins) but is treated with gravity/CIL, so effectively free-milling for gold.
- **Goldex (QC, Agnico Eagle):** An underground gold mine. A 1,800 t/d mill employs crushing, grinding, gravity recovery and CIL, plus flotation for sulfide. Roughly two-thirds of the gold is recovered by gravity and smelted on-site to doré, with the rest recovered as a gold–pyrite concentrate shipped to a central mill [7]. The ore (Greenstone-hosted lode gold) is free-milling (gravity-rich).
- **Hemlo (Williams, ON, Barrick):** Formerly surface, now underground gold mining at Williams. Processing (up to 10,000 t/d) uses crushing, grinding, cyanide leaching, CIP and electrowinning [8]. Most gold is recovered to doré on-site (gravity/CIP), with any sulphide concentrate typically shipped out. Hemlo ore (shear zone gold) is essentially free-milling.
- **Island Gold (ON, Alamos/Magino):** An underground gold mine (long-hole stoping) combined with the new Magino open pit (project). Ore from both is treated in a shared mill network: the Magino mill (10,000 t/d) plus the old Island/Kremzar mill (1,200 t/d). These use SAG/Ball mills, gravity and CIP for gold recovery [9]. The carbon-in-pulp plant pours doré bars. The granodiorite-hosted gold ore is free-milling.
- **Kiena (QC, Wesdome):** An underground gold mine. A 2,040 t/d mill (cyanide CIL) processes the ore, producing doré bars (electrowinning) and gravity concentrates [10]. The orogenic lode gold ore is treated as free-milling, with no refractory step.
- **Lamaque (QC, Eldorado):** An underground gold complex (Triangle and Ormaque deposits) in Val-d'Or. All ore is trucked 2–3 km to the Sigma Mill for processing (conventional crushing, grinding, gravity and CIP).

The Sigma Mill pours gold-silver doré. Lamaque ore is archean lode gold (free-milling orogenic type) [11].

- **LaRonde (QC, Agnico Eagle):** A deep underground gold mine (LaRonde & LZ5 zones). The mill includes flotation (for copper/zinc) plus a CIP circuit with a two-stage roast for refractory arsenopyrite. The plant produces gold–silver doré, and zinc and copper concentrates (with precious metal credits) [12]. LaRonde ore is high-sulfide (arsenical) and refractory (hence the roaster).
- **Macassa (ON, Kirkland Lake):** A very high-grade underground gold mine (Longhole & cut-and-fill). The 1,000–1,300 t/d plant uses crushing, fine grinding (tower mill), intensive cyanidation and CIP (the IG – intensive grind – process, Zadra zinc precipitation) [13]. The melt produces doré bars (70–85% Au) [13]. Ore is hard gold-bearing quartz sulphide; it is treated via autoclave/CIP (Zadra) but can be viewed as partially refractory, given the IG step.
- **Meliadine (NU, Agnico Eagle):** A gold mine with multiple open pits and underground zones. A 4,000 t/d mill uses crushing, grinding, gravity concentration and cyanidation with CIL [14], followed by carbon stripping and tailings filtration. The plant pours doré bars. The greenstone-hosted ore is free-milling, designed for conventional recovery.
- **Musselwhite (ON, Newmont):** An underground gold mine (long-hole & cut-and-fill). Ore is processed in a ~1.5 Mtpa mill with crushing, grinding, gravity recovery and CIP (Merrill-Crowe) to doré (no flotation). Approximately 50–60% of gold is recovered by gravity [15] (free-milling). (Historical data indicate 250–300 koz/yr at ~$300/oz, reflecting free-milling ore.)
- **Rainy River (ON, New Gold):** An open pit and ramp-access underground mine. A 12,000 t/d plant uses two-stage crushing, grinding, gravity separation, cyanide leaching, CIP and electrowinning. The recovered gold (~40% Au in dore bars, balance Ag) is poured as doré [16]. The Archean ore (greenstone gold) is free-milling; no pressure oxidation is used.
- **Red Lake (ON, Newmont):** A legendary high-grade underground gold complex (Campbell and Red Lake mines). A 1,250 t/d mill (updated 2000) uses two-stage crushing, gravity concentration and carbon-in-pulp with electrowinning (and on some ore, autoclave) [17]. Gold is produced as doré. The ore is extremely high-grade free gold (with some refractory zones treated by autoclave). About 98% of gold is recoverable.
- **Westwood–Doyon (QC, IAMGOLD):** The Westwood Mine (UG) plus Grand Duc pit are processed at the nearby Doyon mill. The plant (2,100 t/d) uses crushing, grinding, gravity recovery and CIL [18]. Gold is recovered to doré bars. The deposits are orogenic lode and VMS, treated as free-milling. The site mix is ~70% underground (Westwood) and ~30% open pit (Grand Duc) [18].
- **Young-Davidson (ON, Alamos):** A large underground gold mine. The 4,400 t/d CIP mill (expanded in 2020) uses conventional crushing, SAG/ball milling and carbon-in-pulp (electrowinning) to produce doré. (Alamos notes it as one of Canada's largest underground gold mines [19].) The ore is orogenic gold and treated as free-milling.
- **Meadowbank Complex (NU, Agnico Eagle):** Includes the Meadowbank open pit and the satellite Amaruq (open pit + UG). A ~4,000 t/d plant employs crushing, grinding, gravity and CIL [20]. Gravity concentrates and gold-plated cathodes (from electrowinning) are melted to doré. The ore is typical Archean greenstone gold (free-milling).
- **Porcupine Complex (ON, Newmont):** (Dome, Campbell, etc.) Mostly underground gold. Ore is trucked to the Timmins mill. The processing (crush-grind + CIP) produces doré bars. The Abitibi orogenic lode ore is free-milling with gravity recovery.
- **Timmins Operation (ON, Newmont):** (Hoyle Pond UG, Holt pit). The Timmins mill (~4,700 t/d) crushes, grinds and leaches ore (CIP) to doré. (Hoyle Pond is deep underground; Holt is open pit.) The local orogenic gold ore is free-milling.
- **Seabee Gold Operation (SK, SSR Mining):** Underground (Santoy & historical Seabee shafts). A 1,800 t/d mill (built 1991) uses crushing, grinding and CIP. All recovered gold is poured as doré [21]. The quartz-vein ore is high-grade and free-milling.

Copper/Metals Mines:
- **Copper Mountain (BC, Copper Mountain Mining):** A large open-pit Cu-Au mine. A 45,000 t/d processing plant (crushing, SAG/ball, flotation) produces copper concentrates (with gold/silver credits) [22]. No on-site smelting; concentrate is shipped out. The ore is a porphyry (sulfide) copper–gold deposit.
- **Gibraltar (BC, Taseko Mines):** One of Canada's largest open-pit copper mines. An 85,000 t/d mill (crush-grind, flotation) produces copper and molybdenum concentrates [23]. (Past production: ~110 M lbs Cu/year.) No dore; ore is porphyry sulfide.
- **Mount Milligan (BC, Centerra):** An open-pit copper-gold porphyry. A 60,000 t/d mill (crushing, grinding, flotation) yields copper concentrate (with gold byproduct) [24]. (Centerra reports ~168 koz Au, 54 M lbs Cu in 2024.) No on-site smelting.
- **New Afton (BC, New Gold/Gold Resource Corp):** An underground block-cave copper–gold mine (former Afton pit). A 2,700 t/d mill (grinding, flotation) produces copper–gold concentrate [25]. The concentrate (with Au/Ag credits) is shipped to smelters. The ore is porphyry (Cu-Au sulphide).
- **Red Chris (BC, Newcrest/Newmont):** An open-pit copper–gold porphyry. A ~30,000 t/d plant (installed in 2015) crushes, mills and floats to produce copper–gold concentrate. (Transition to block-cave UG planned [26].) No dore production.
- **Keno Hill Silver (YT, Alexco):** Underground silver–lead–zinc. A 400 t/d mill (crush-grind, flotation) produces silver–lead–zinc concentrates [27]. (Primary metal: silver. No doré.)
- **Lac des Iles (ON, Impala Canada):** Ontario's largest palladium (PGM) mine. Both open pit (past) and underground (long-hole) operations feed a concentrator (rated ~3.6 Mtpa) [28]. The mill (crush-grind, flotation) produces a bulk PGM–Ni–Cu concentrate. No doré.
- **Carol Lake (NL, IOC):** An integrated open-pit iron mine. Five pits feed a concentrator and pellet plant [29]. The ~10–12 Mtpa concentrator processes magnetite ore (to pellets). (No relevance to gold.)
- **Scully (NL, Tacora):** A large open-pit iron ore operation (Wabush area). It feeds a pellet plant. (Iron ore is magnetite; processed to concentrate/pellets.)

Snow Lake (Manitoba, Hudbay): This mining camp includes the Lalor copper–zinc–gold base-metal mine and the New Britannia gold mine. Lalor (open pit/UG) is a Cu-Zn-Pb-Au deposit: ore is milled at Stall concentrator to base-metal concentrates (with significant gold recovery) and also processed at the refurbished New Britannia mill. The New Britannia mill (~1,500–1,800 t/d) handles gold-rich ore, with crushing, grinding and CIP to produce gold doré. (Lalor's gold was ~1 Moz by 2024 [30].) Gold ore in Snow Lake is free-milling (oxide or sulphide); copper/zinc ore is sulfide and goes to concentrates.

Ore Types: Most listed gold mines are free-milling (orogenic/quartz-vein gold) processed by gravity + cyanide. A few (LaRonde, Macassa) involve refractory sulfide (arsenopyrite/pyrite) requiring roasting or autoclave pre-treatment. Copper/porphyry ores are sulphide and sent to flotation concentrators (no doré), while Carot Lake and Scully are magnetite iron ores (concentrated/pelletized) [29]. Silver–lead–zinc ores (Keno Hill) are also sulphide, sent to flotation.

Sources: Authoritative mine and company reports were used for each mine's methods and processing. For example, Newmont's site notes Brucejack's underground mill producing doré; Agnico's pages describe Malartic, LaRonde, Macassa, etc. [12] [13]; IAMGOLD describes Westwood/Doyon processing [18]; Barrick details Hemlo's CIP plant [8]; SSR Mining confirms Seabee's doré plant [21]; and various technical reports/ mining databases give similar details. All information above is drawn from these sources [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [24] [25] [16] [17] [18] [20] [21] [22] [23] [28] [27] [29].

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